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The Editors
Empirical Software Engineering
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Dear Editors,

I submit for your consideration the manuscript **"What Should a Graph Node Say? A Controlled Study of Node Descriptions for Cross-Project Vulnerability Detection"** as a research article.

What the paper does. Graph-based vulnerability detectors report good accuracy on projects seen in training and much worse accuracy on projects they have not seen. I ask how much of that failure depends on what each node of the program graph tells the network. With the graph, network and training protocol held fixed, I compare four node descriptions (identifier text, text with identifiers renamed, the parser's grammar kind, and a hand-built operation label) on uniform random samples of BigVul and DiverseVul, using project-disjoint folds, a cluster bootstrap over held-out projects, direct paired tests, and controls for function size and for single dominant projects.

Main findings, including the negative ones. Text embeddings learned from scratch transfer close to chance. Renaming identifiers or using grammar kinds improves AUC substantially, and the two do about equally well. The hand-built operation label adds nothing to the grammar kind, which I explain with an exact information-theoretic argument. Much of the gain disappears when function size is held fixed, and a one-line function-size rule has a higher mean AUC than every network, while on BigVul a classifier that sees only formatting (trailing whitespace is present in 61% of vulnerable and 2% of benign functions) reaches AUC 0.896. A fine-tuned CodeBERT falls from 0.922 on raw source to 0.728 once formatting is removed; it then beats the graph models by a small margin but is not distinguishable from the size rule. I report these results at face value and recommend that studies of cross-project detectors include a function-size baseline and a vocabulary-free baseline.

Fit with the journal. The paper is an empirical study with a strong emphasis on evaluation methodology: controlled comparison, explicit threats to validity, annotation agreement for the labelling function, and every number generated by released scripts from stored predictions. It falls under "verification and validation" and "applications of artificial intelligence techniques to software engineering" in the journal's aims and scope.

Transparency. An earlier version of this work, with a different (federated-learning) framing, was submitted to another journal and not accepted; a second version with the present representation framing was also reviewed elsewhere and not accepted. The present manuscript responds to those reviews with new experiments (random samples, ten matched folds, additional baselines, a Transformer and a pretrained model, ablations, statistical tests, function-size controls and a redone annotation protocol) and withdraws the claims that the new evidence did not support. The manuscript is not under consideration elsewhere. All code, annotation sheets and per-fold predictions are available at `https://github.com/vinhqdang/vulmorph-fed`. Limitations that I could not remove (a short fine-tuning schedule for the pretrained model, two language-model annotators and no human second annotator, samples of about 10,000 functions) are stated in the paper.

Thank you for your consideration.

Yours sincerely,

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